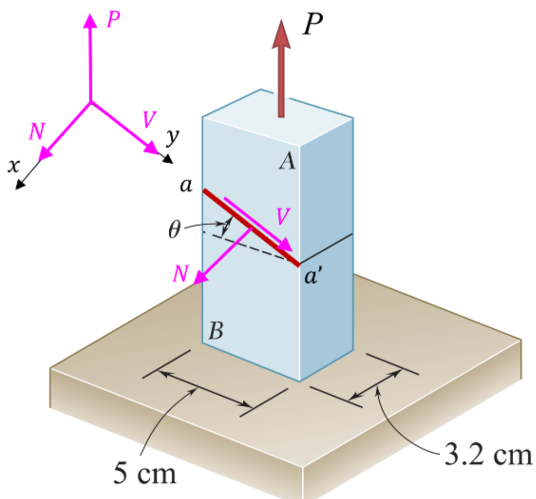
Final Exam for Solid Mechanics (EMCH260 Section 002, Fall 2021)
9:00 – 11:30 AM, Wednesday, Dec. 8th 2021

Name (First/Last): ______________________________ **Score:** ______________/100'

1. The two portions of member A and B are glued together along a plane aa' forming an angle $\theta = 30^\circ$ with respect to the horizontal. The lateral size of the member is 5 cm × 3.2 cm. Knowing that the allowable stress for the glued joint is 50 MPa in tension and 25 MPa in shear, determine the maximum possible external force (P) so that the glued joint will not break. (**10'** in total)



Solutions:

Assuming the normal (tensile) force and shear force at the glued joint as N (in x-direction) and V, respectively, and applying the free body diagram to the member A, we get

$\sum F_x = 0 \quad \rightarrow \quad N = P \cdot \cos\theta$

$\sum F_y = 0 \quad \rightarrow \quad V = P \cdot \sin\theta$

Then, the normal and shear stress at the joint plane is (note the cross-section of the joint)

$\sigma = \frac{N}{A} = P \cdot \frac{\cos\theta}{\frac{A_0}{\cos\theta}} = \frac{P}{A_0}\cos^2\theta,$ where $A_0 = 0.05 \times 0.032 = 0.0016 \text{ m}^2$

$\tau = \frac{V}{A} = P \cdot \frac{\sin\theta}{\frac{A_0}{\cos\theta}} = \frac{P}{A_0}\sin\theta\cos\theta$

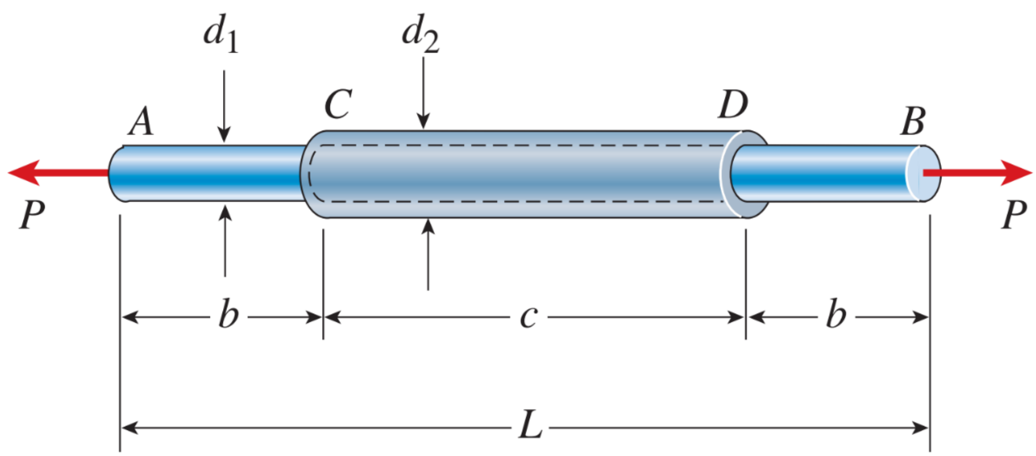
To keep both the maximum normal and shear stress below the ultimate stress, we have

$\frac{P}{A_0}\cos^2\theta \leq 50 \text{ MPa} \quad \rightarrow \quad P \leq 106.67 \text{ kN}$

$\frac{P}{A_0}\sin\theta\cos\theta \leq 25 \text{ MPa} \quad \rightarrow \quad P \leq 92.38 \text{ kN}$

Therefore, the maximum possible external force is $\boxed{\boldsymbol{P \leq 92.38 \text{ kN}}}$

2. A plastic rod AB of length $L = 0.5$ m has a diameter $d_1 = 30$ mm (see figure). A plastic sleeve CD of length $c = 0.3$ m and outer diameter $d_2 = 45$ mm is securely bonded to the rod so that no slippage can occur between the rod and the sleeve ($b = 0.1$ m). The rod is made of an acrylic with modulus of elasticity $E_1 = 3.1$ GPa and Poisson's ratio $\nu = 0.3$, and the sleeve is made of a polyamide with $E_2 = 2.5$ GPa. Determine the elongation (δ) of the rod AB and the new diameter ($d_1{}'$) of the AC or DB segment, when the rod is pulled by axial forces $P = 12$ kN. Note: take $\pi = 3.1415926$. (**15'** in total)



Solutions:

For AC and DB segments, the problem is determinate.

Elongation of AC and DB segments is:

$$\delta_{AC} = \frac{PL_{AC}}{E_{AC}A_{AC}} = \frac{12000\times0.1}{3.1\times10^9\times\frac{\pi}{4}\times0.03^2} = 5.4763\times10^{-4}\text{ m} \qquad (1)$$

$$\delta_{DB} = \delta_{AC} = 5.4763\times10^{-4}\text{ m} \qquad (2)$$

For CD segment, the problem is statically indeterminate.

Assuming the resultant tensile force for the rod and sleeve is P_{rd} and P_{sl}, respectively, the free body diagram yields:

$$P_{rd} + P_{sl} = P = 12000 \qquad (3)$$

The Compatibility Condition is that the elongation of the rod (only the CD segment) and sleeve is the same, i.e.

$$\delta_{rd} = \delta_{sl}$$

$$\rightarrow \quad \frac{P_{rd}L_{rd}}{E_{rd}A_{rd}} = \frac{P_{sl}L_{sl}}{E_{sl}A_{sl}}$$

Taking $L_{rd} = L_{sl}$, $E_{rd} = 3.1 \times 10^9$, $E_{sl} = 2.5 \times 10^9$, $A_{rd} = \frac{\pi}{4} \times 0.03^2$, $A_{sl} = \frac{\pi}{4} \times (0.045^2 - 0.03^2)$, we get:

$$\frac{P_{rd}}{3.1 \times 0.03^2} = \frac{P_{sl}}{2.5 \times (0.045^2 - 0.03^2)}$$

$$P_{rd} = 0.992 P_{sl} \qquad (4)$$

Substituting Eq. (4) into Eq. (3) we get:

$$P_{rd} = 5976 \text{ N}$$

$$P_{sl} = 6024 \text{ N}$$

Then,

$$\delta_{rd} = \frac{P_{rd} L_{rd}}{E_{rd} A_{rd}} = \frac{5976 \times 0.3}{3.1 \times 10^9 \times \frac{\pi}{4} \times 0.03^2} = 8.1816 \times 10^{-4} \text{ m}$$

Finally, the total elongation of the rod is:

$$\delta = \delta_{AC} + \delta_{rd} + \delta_{DB} = 19.1342 \times 10^{-4} \text{ m} = \boxed{\mathbf{1.913\ mm}}$$

Longitudinal strain of AC or DB segment is

$$\varepsilon_{\text{long}} = \frac{P}{E_{AC} A_{AC}} = \frac{12000}{3.1 \times 10^9 \times \frac{\pi}{4} \times 0.03^2} = 0.0054763 \text{ mm/mm}$$

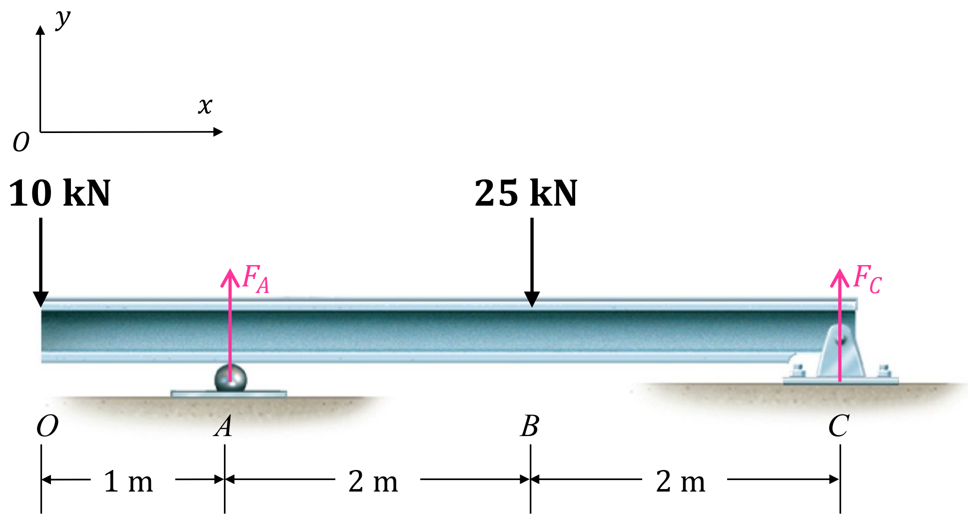
According to the definition of the Poisson's ratio ($\nu = -\frac{\varepsilon_{\text{lat}}}{\varepsilon_{\text{long}}}$), the strain in the lateral direction is

$$\varepsilon_{\text{lat}} = -\nu \cdot \varepsilon_{\text{long}} = -0.3 \times 0.0054763 = -0.00164289 \text{ mm/mm}$$

Then, with the definition of lateral strain, we obtain the new diameter of the *AC* or *DB* segment as

$$d_1' = (1 + \varepsilon_{\text{lat}}) \cdot d_1 = (1 - 0.00164289) \times 30 = \boxed{\mathbf{29.95\ mm}}$$

3. Determine the shear and bending moment diagrams of the beam with external loading shown below, and also determine the absolute maximum normal stress in the beam. The beam has a uniform square cross-section with lateral size $a = 0.1$ m. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (**15'** in total)



Solutions:

Graphical method to calculate the shear and moment diagrams.

Step 1: Support Reactions

Applying Free-Body-Diagram to the entire beam, the force equation of equilibrium gives:

$\sum F_y = 0 \quad \rightarrow \quad F_A + F_C = 10 + 25 = 35 \text{ kN} \qquad (1)$

$\sum M_C = 0 \quad \rightarrow \quad 10 \times 5 + 25 \times 2 = F_A \times 4 \qquad (2)$

Solving Eqs. (1) and (2) yields:

$F_A = 25 \text{ kN}$

$F_C = 10 \text{ kN}$

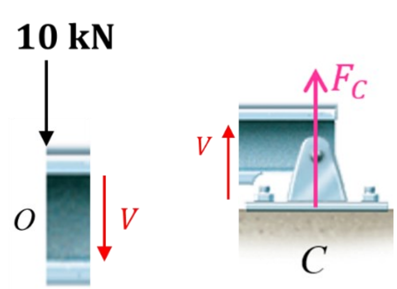
Step 2: Shear Diagram

a) Using the FBD and sign convention for shear force, determine the shear at the left end (O point):

$x = 0, \ V = -10 \text{ kN}$

At the right end (C point):

$x = 5 \text{ m}, \ V = -F_C = -10 \text{ kN}$



b) Determine **V** diagram according to $\frac{dV}{dx} = w$ for $\underline{OA}$ region:

$$\frac{dV}{dx} = w = 0$$

$\rightarrow V = C_1$

The constant C_1 can be determined by applying the "boundary" condition ($x = 0,\ \ V = -10$ kN). Thus, $C_1 = -10$ and $V = -10$ kN $(0 \le x \le 1$ m).

c) At point A, due to the support reaction force F_A (upward), the **V** increases (upward jump) by $F_A = +25$ kN. Then, the **V** jumps from -10 kN to $+15$ kN.

d) Determine **V** diagram according to $\frac{dV}{dx} = w$ for $\underline{AB}$ region:

$$\frac{dV}{dx} = w = 0$$

$$\rightarrow\ V = C_2$$

The constant C_2 can be determined by the "**<u>continuity</u>**" condition at the last point of OA region after jump ($x = 1$ m, $V = +15$ kN). Thus, $C_2 = +15$ kN and $V = +15$ kN $(1 \le x \le 3$ m).

d) At point B, due to the external concentrated force 25 kN (downward), the **V** decreases (downward jump) by 25 kN. Then, the **V** jumps from $+15$ kN to -10 kN.

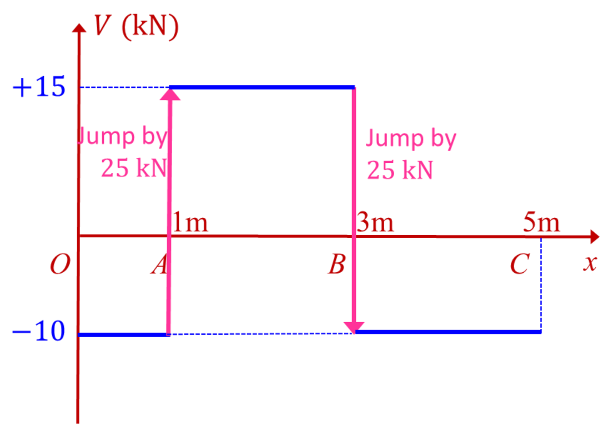
e) Determine **V** diagram according to $\frac{dV}{dx} = w$ for $\underline{BC}$ region:

$$\frac{dV}{dx} = w = 0$$

$$\rightarrow\ V = C_3$$

The constant C_3 can be determined by the "**<u>continuity</u>**" condition at the last point of AB region after jump ($x = 3$ m, $V = -10$ kN). Thus, $C_3 = -10$ kN and $V = -10$ kN $(3 \le x \le 5$ m).

The shear diagram can be plotted as shown.



Verify:

At right end (point C, $x = 5$ m), the internal resultant shear force is -10 kN, which is consistent with our previous boundary condition.

<u>Step 3: Moment Diagram</u>

a) Determine **M** at left and right end:

$$x = 0,\ \ M = 0$$

$$x = 5\text{ m},\ \ M = 0$$

b) Determine **M** diagram according to $\frac{dM}{dx} = V$ for the $\underline{OA}$ region:

$$\frac{dM}{dx} = V = -10$$

$$\rightarrow\ M = -10x + C_4$$

The constant C_4 can be determined by applying the "boundary" condition ($x = 0,\ M = 0$). Thus, $C_4 = 0$ and $M = -10x\ \ (0 \le x \le 1\text{ m})$.

c) Determine **M** diagram according to $\frac{dM}{dx} = V$ for <u>*AB*</u> region:

$$\frac{dM}{dx} = V = 15$$

→ $M = 15x + C_5$

The constant C_5 can be determined by applying the "**<u>continuity</u>**" condition at last point *A* of the *OA* region ($x = 1,\ M = -10\text{ kN}\cdot\text{m}$). Thus, $C_5 = -25\text{ kN}\cdot\text{m}$ and $M = 15x - 25\text{ kN}\cdot\text{m}\ \ (1 \le x \le 3\text{ m})$.

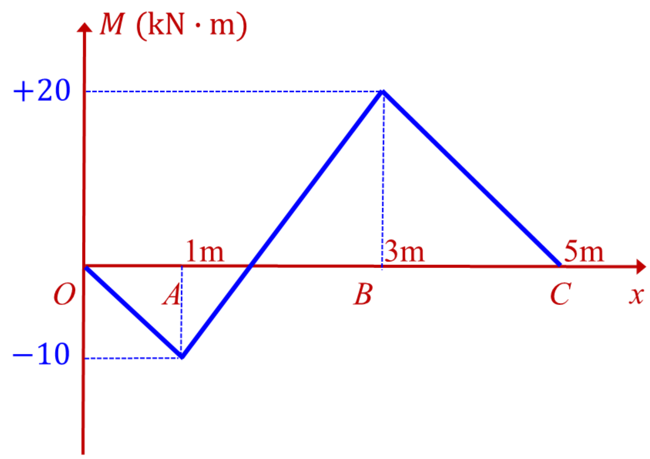
d) Determine **M** diagram according to $\frac{dM}{dx} = V$ for <u>*BC*</u> region:

$$\frac{dM}{dx} = V = -10$$

→ $M = -10x + C_6$

The constant C_6 can be determined by applying the "**<u>continuity</u>**" condition at last point *B* of the *AB* region ($x = 3,\ M = +20\text{ kN}\cdot\text{m}$). Thus, $C_6 = 50\text{ kN}\cdot\text{m}$ and $M = -10x + 50\text{ kN}\cdot\text{m}\ \ (3 \le x \le 5\text{ m})$.

The moment diagram can be plotted as shown.



Verify:

At right end (point *C*, $x = 5$ m), the internal resultant bending moment is 0, which is consistent with our previous boundary condition.

Therefore, the maximum moment throughout the entire beam is $M_{\max} = 20\text{ kN}\cdot\text{m}$ at $x = 3$ m.

<u>Step 4: Section Property</u>

The moment of inertia *I* of sectional area about the neutral axis is (refer to the inside cover of the textbook for rectangular cross-section):

$$I = \frac{1}{12}bh^3 = \frac{1}{12} \times 0.1 \times 0.1^3 = 8.3333 \times 10^{-6}\text{ m}^4$$

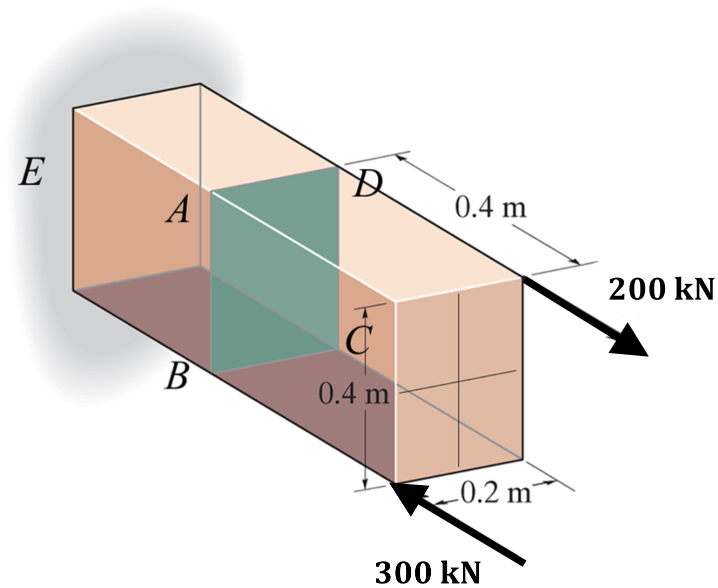
<u>Step 4: Maximum Normal Stress</u>

The absolute maximum normal stress is determined by:

$$\sigma_{\max} = \frac{M_{\max}c}{I}$$

Thus, $\sigma_{\max} = \frac{20000 \times 0.05}{8.3333 \times 10^{-6}} = \boxed{\textbf{120 MPa}}$

4. The brick is subjected to the loads shown. The cross section throughout the entire brick is 0.2 m × 0.4 m. Determine the stress at the corner point *A*, *B*, *C*, and *D*. The cross section *ABCD* is perpendicular to the longitudinal direction of the brick. The end *E* of the rod is fixed. (**10'** in total)



Solutions:

Step 1: Internal Loading

Section the member along the *ABCD* plane, where the stress is to be determined. The *xyz* coordinate system is established with origin at the centroid point of *ABCD*.

The resultant internal normal force and bending moment can be obtained as:

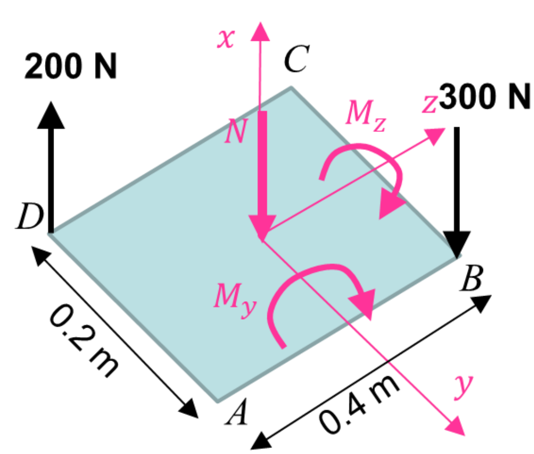
- Normal force in *x*-direction:

$$N = \sum F_x = 200 - 300 = -100 \text{ kN}$$

- Bending moment components:

$$M_y = -200 \times 0.2 - 300 \times 0.2 = -100 \text{ kN} \cdot \text{m}$$

$$M_z = 300 \times 0.1 + 200 \times 0.1 = +50 \text{ kN} \cdot \text{m}$$



Step 2: Cross-section Property

$$A = 0.2 \times 0.4 = 0.08 \text{ m}^2$$

$$I_y = \frac{1}{12} \times 0.2 \times 0.4^3 = 1.0667 \times 10^{-3} \text{ m}^4$$

$$I_z = \frac{1}{12} \times 0.4 \times 0.2^3 = 2.6667 \times 10^{-4}\ \mathrm{m}^4$$

Step 3: Stress Components

The normal stress developed is the combination of axial and bending stress.

$$\sigma_N = \frac{N}{A}, \quad \sigma|_{M_y} = -\frac{M_y \cdot \boldsymbol{z}}{I_y}, \quad \sigma|_{M_z} = -\frac{M_z \cdot \boldsymbol{y}}{I_z}$$

Thus,

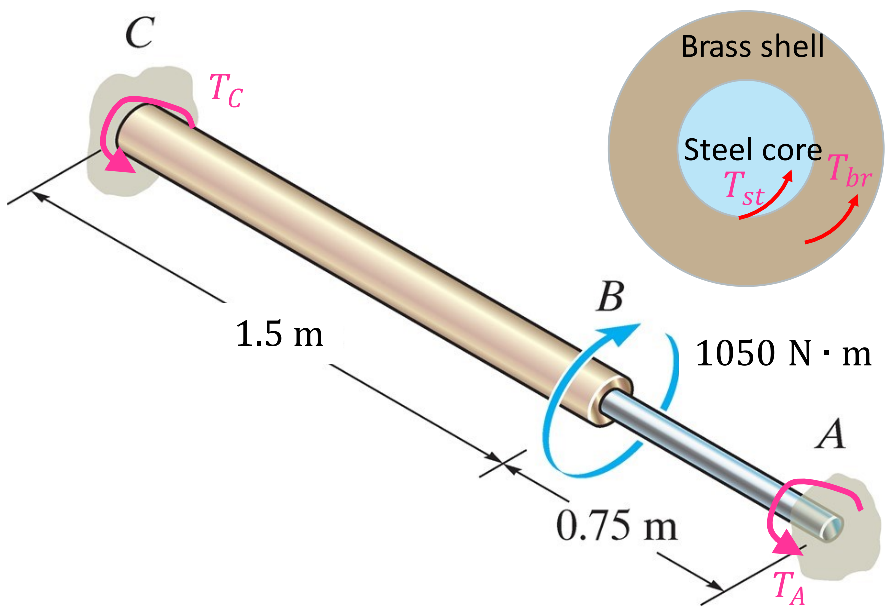
$$\sigma_A = -\frac{100}{0.08} + \frac{100\times0.2}{1.0667\times10^{-3}} - \frac{50\times0.1}{2.6667\times10^{-4}} = \boxed{\mathbf{-1.25\ MPa}}$$ (compressive)

$$\sigma_B = -\frac{100}{0.08} - \frac{100\times0.2}{1.0667\times10^{-3}} - \frac{50\times0.1}{2.6667\times10^{-4}} = \boxed{\mathbf{-38.75\ MPa}}$$ (compressive)

$$\sigma_C = -\frac{100}{0.08} - \frac{100\times0.2}{1.0667\times10^{-3}} + \frac{50\times0.1}{2.6667\times10^{-4}} = \boxed{\mathbf{-1.25\ MPa}}$$ (compressive)

$$\sigma_D = -\frac{100}{0.08} + \frac{100\times0.2}{1.0667\times10^{-3}} + \frac{50\times0.1}{2.6667\times10^{-4}} = \boxed{\mathbf{36.25\ MPa}}$$ (tensile)

5. A shaft is made from two segments: AB is pure steel and BC is **core-shell** (core: steel, shell: brass). The entire shaft is fixed at its two ends and subjected to a torque of $T = 1050\ \mathrm{N \cdot m}$ at point B. The AB portion has a diameter of 20 mm. The inner diameter and outer diameter of the core-shell part is 20 mm and 40 mm, respectively. The shear modulus of elasticity of steel and brass is 80 GPa and 40 GPa, respectively. Determine the reactions at the two ends A and C. (**15'** in total)



Solutions:

Assume the reactions at two ends as T_A and T_C. Apply Free-Body-Diagram to the entire shaft:

$$T_A + T_C = 1050 \qquad (1)$$

Since the BC part is core-shell, we assume the internal resultant moment portion from steel core and brass shell as T_{st} and T_{br}, respectively. Then, we have:

$$T_{st} + T_{br} = T_C \qquad (2)$$

Compatibility conditions:

For the core-shell part:

$$\phi_{st} = \phi_{st} = \phi_{BC} \qquad (3)$$

Using the angle of twist formula, we get:

$$\frac{T_{st} L_{BC}}{J_{st} G_{st}} = \frac{T_{br} L_{BC}}{J_{br} G_{br}}$$

Using the geometry of core-shell shaft, we can simply the above equation:

$$\frac{T_{st}}{\frac{\pi}{2}\cdot 0.01^4\cdot 80\times 10^9} = \frac{T_{br}}{\frac{\pi}{2}\cdot (0.02^4 - 0.01^4)\cdot 40\times 10^9}$$

→ $T_{br} = 7.5T_{st}$ (4)

For the entire shaft:

$\phi_{AC} = 0$ → $\phi_{AB} + \phi_{BC} = 0$

Using the angle of twist formula and Eq. (3) (we use steel part to calculate ϕ_{BC}), we get:

$$\frac{T_{AB}L_{AB}}{J_{st}G_{st}} - \frac{T_{st}L_{BC}}{J_{st}G_{st}} = 0$$

Since $T_{AB} = T_A$ (use Free-Body-Diagram for segment *AB*), then we have:

$T_A = 2T_{st}$ (5)

Combining Eqs. (2) + (4) yields:

$T_C = 8.5T_{st}$ (6)

Substituting Eqs. (5) and (6) into Eq. (1) we have:

$2T_{st} + 8.5T_{st} = 1050$

→ $T_{st} = 100\ \mathrm{N\cdot m}$

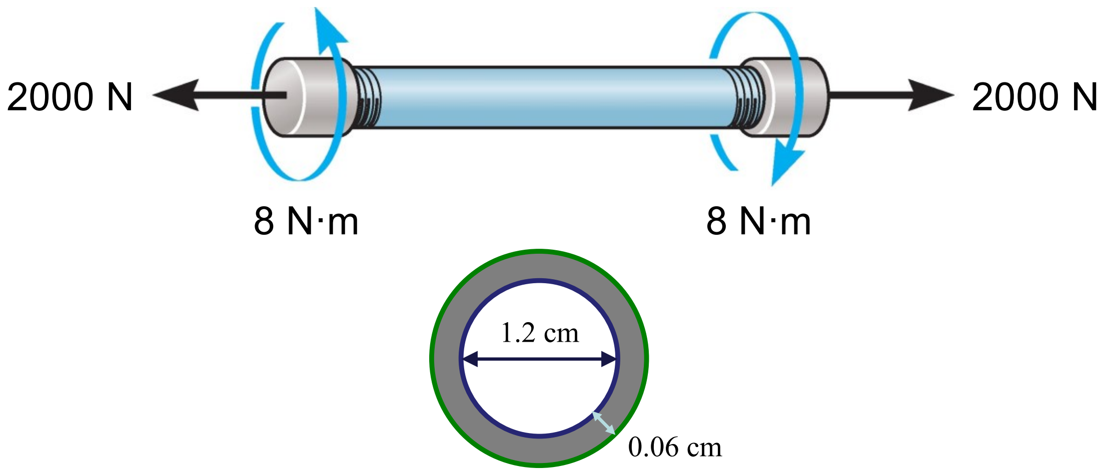
Finally, according to Eqs. (5) and (1), we get:

$T_A = 2T_{st} = \boxed{\mathbf{200\ N\cdot m}}$

$T_C = 1050 - T_A = \boxed{\mathbf{805\ N\cdot m}}$

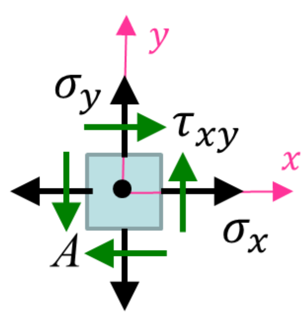
6. The thin-walled pipe has an inner diameter of 1.2 cm and a thickness of 0.06 cm. If it is subjected to an internal pressure of 4 MPa and the axial tension and torsional loadings shown, determine the principal stress and the absolute maximum shear stress for a point on the surface of the pipe. (**15'** in total)

Hint: the cross-section of the vessel can be regarded as a hollow tube and the cross-section area can be calculated as $\frac{\pi}{4}(0.0132^2 - 0.012^2)$. You can do the same treatment for the polar moment inertia.



Solutions:

In this gas filled vessel, in addition to the two non-equal normal stresses in the two perpendicular directions (one in the longitudinal direction and the other in the hoop direction), there are shear stress induced by the torque and normal stress induced by the tension force.



Cross-sectional area:

$$A = \frac{\pi}{4}(0.0132^2 - 0.012^2) = 2.375 \times 10^{-5}\ \mathrm{m}^2$$

Polar moment of inertia:

$$J = \frac{\pi}{2}(0.0066^4 - 0.006^4) = 9.448 \times 10^{-10}\ \mathrm{m}^4$$

For a point on the surface of the pipe, the stress state is plane stress.

Establish *Oxy* coordinate system with *x*-axis long the longitudinal direction and *y*-axis along the hoop direction, and then we have the stress components:

$$\sigma_x = \sigma_n + \frac{F}{A} = \frac{pr}{2t} + \frac{F}{A} = \frac{4 \times 10^6 \times 0.6}{2 \times 0.06} + \frac{2000}{2.375 \times 10^{-5}} = 104.21\ \mathrm{MPa}$$

$$\sigma_y = \sigma_a = \frac{pr}{t} = \frac{4 \times 10^6 \times 0.6}{0.06} = 40\ \mathrm{MPa}$$

$$\tau_{xy} = \frac{Tc}{J} = \frac{8 \times 0.0066}{9.448 \times 10^{-10}} = 55.88 \text{ MPa}$$

Given $\sigma_x = 104.21$ MPa, $\sigma_y = 40$ MPa, $\tau_{xy} = 55.88$ MPa, the center of the Mohr's circle C is located at the point $C(72.105, 0)$ and the reference point $A(104.21, 55.88)$ are labeled. Then, the radius of the Mohr's circle is $R = \sqrt{(104.21 - 72.105)^2 + 55.88^2} = 64.45$, and the Mohr's circle is constructed.

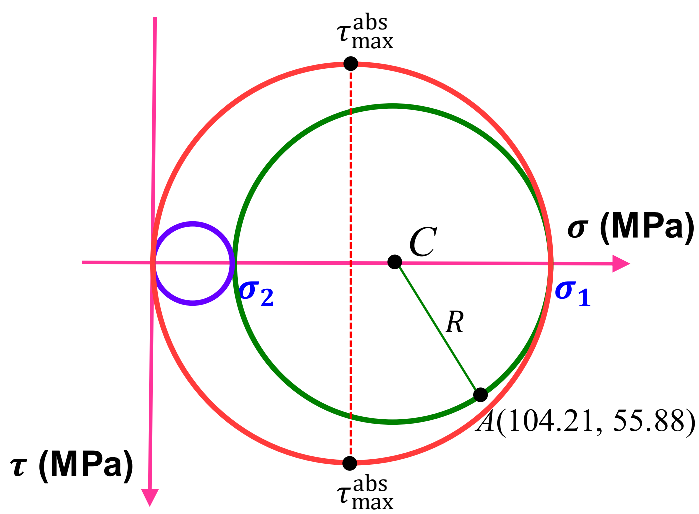
Based on the Mohr's circle, the principal stresses are:

$$\sigma_1 = 72.105 + 64.45 = \boxed{\mathbf{136.6\ MPa}}$$

$$\sigma_2 = 72.105 - 64.45 = \boxed{\mathbf{7.66\ MPa}}$$

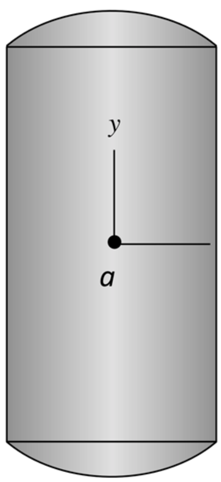
Drawing three Mohr's circle to determine the absolute maximum shear stress:

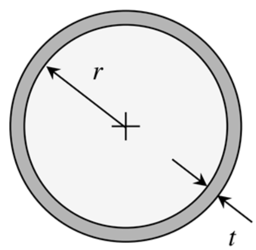
$$\tau_{max}^{abs} = \frac{\sigma_1}{2} = \boxed{\mathbf{68.3\ MPa}}$$



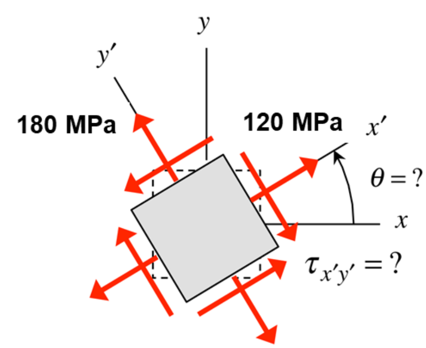
7. The closed thin-walled tank has an inner radius of r and wall thickness t and contains a gas of pressure of p. The state of stress at point "*a*" on the surface of the tank is represented by stress components with an unknown stress element rotation angle of θ. Note that, the two normal stress components for $x'y'$ system (note: not *xy* system!) are known: $\sigma_{x'} = 120$ MPa and $\sigma_{y'} = 180$ MPa, but the shear stress component ($\tau_{x'y'}$) is unknown. Determine: (**20'** in total)

(1) the principal stresses at point *a* ; (**10'**)
(2) $\tau_{x'y'}$; (**5'**)
(3) rotation angle of θ . (**5'**)





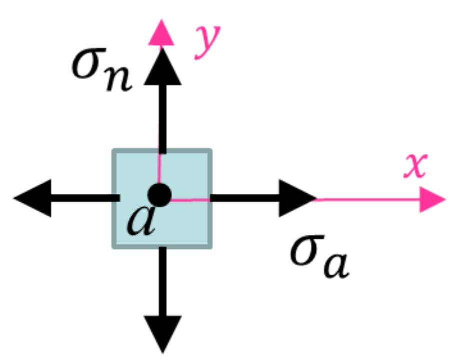
tank cross section



stress state at point a

Solutions:

(1) For point *a*, the state of stress in the original *Oxy* system can be drawn as shown in the right figure. The longitudinal (*y*-direction) normal stress is denoted as σ_n, while the normal stress in the hoop (*x*-) direction is denoted as σ_a. Note that, there is relationship between these two stresses: $\sigma_a = 2\sigma_n$.



Since there is no shear stress in this state of stress, the two stresses are the principal stresses! Then, the Mohr's circle can be immediately drawn, with principal stresses

$$\sigma_{avg} = \frac{\sigma_a + \sigma_n}{2} = 1.5\sigma_n = \frac{\sigma_{x'} + \sigma_{y'}}{2} = 150 \text{ MPa}$$

$$\boxed{\begin{aligned} \sigma_1 &= \sigma_a = 2\sigma_n = \mathbf{200\ MPa} \\ \sigma_2 &= \sigma_n = \mathbf{100\ MPa} \end{aligned}}$$

Note that, both σ_n and σ_a should be positive (i.e. they have same sign).

(2) The radius of the Mohr's circle is:

$$R = \frac{\sigma_1 - \sigma_2}{2} = \frac{200 - 100}{2} = 50 \text{ MPa}$$

$\tau_{x'y'}$ can then be determined by the radius of the Mohr's circle:

$$R^2 = (\frac{\sigma_{x'} - \sigma_{y'}}{2})^2 + (\tau_{x'y'})^2$$

With $\sigma_{x'} = 120$ MPa and $\sigma_{y'} = 180$ MPa, we get

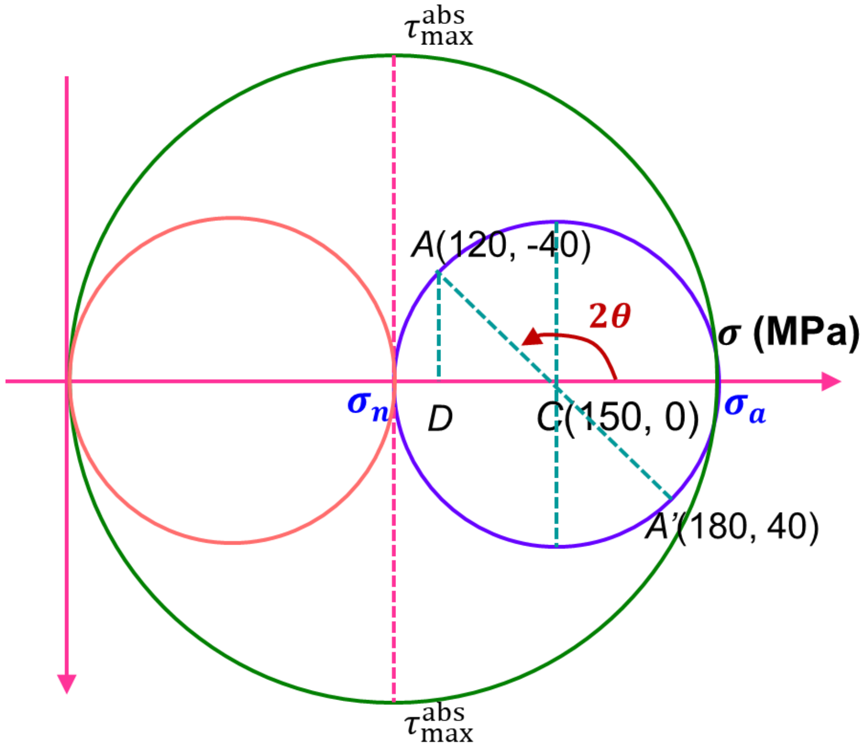
$$\boxed{\boldsymbol{\tau_{x'y'} = -40 \text{ MPa}}}$$

(3) The A point in the Mohr's circle corresponds to the $\sigma_{x'}$ component in the $x'y'$ system. The σ_a point in the Mohr's circle corresponds to the σ_a component in the original xy system. The rotational angle θ can be solved by knowing the angle $\angle ACD$ in the triangle ACD. Note that there is a factor of 2 in the Mohr's circle when you rotate original *xy*-axis to *x'y'*-axis by a counterclockwise angle θ.

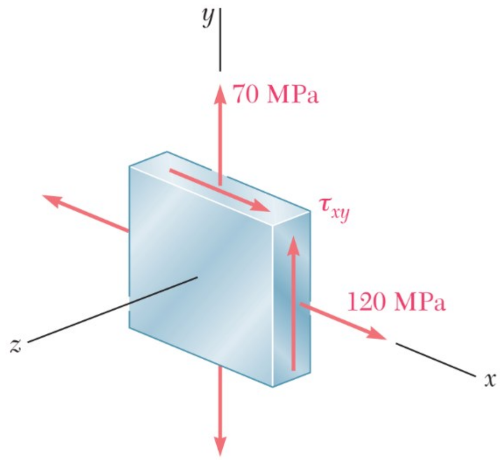
$$\angle ACD = \arctan\frac{40}{150 - 120} = 53.13°$$

Thus, $2\theta = 180° - \angle ACD = 126.87°$,

$$\boxed{\boldsymbol{\theta = 63.44°}}$$



***[Bonus Problem 1] For a plane stress state shown in the figure, if the <u>absolute maximum</u> shear stress is 130 MPa, determine the possible principal stresses in the *xy*-plane. (**10'** in total)



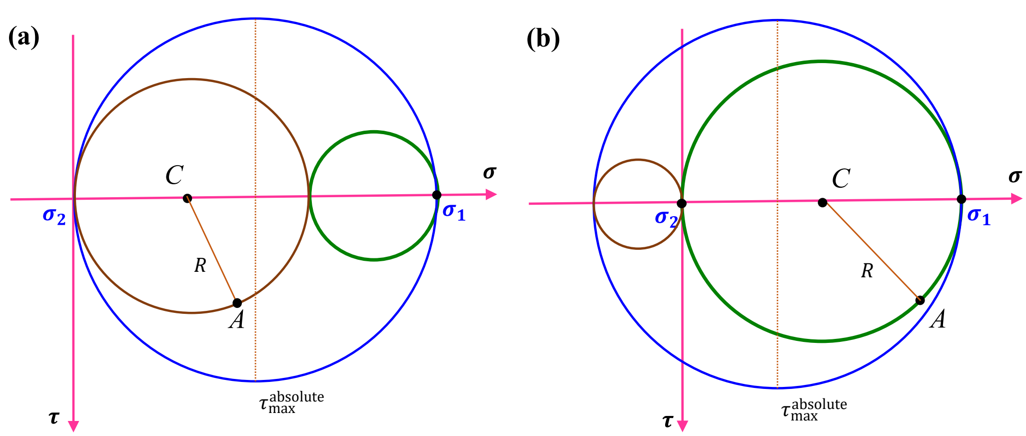
<u>Solutions:</u>

From the stress state, we know the two components of plane stresses: $\sigma_x = 120$ MPa, $\sigma_y = 70$ MPa.

Then, for the Mohr's circle the center C should be $C\left(\frac{\sigma_x+\sigma_y}{2}, 0\right) = C(95, 0)$

The reference point $A(\sigma_x, \tau_{xy})$ cannot be determined since we do not know τ_{xy}.

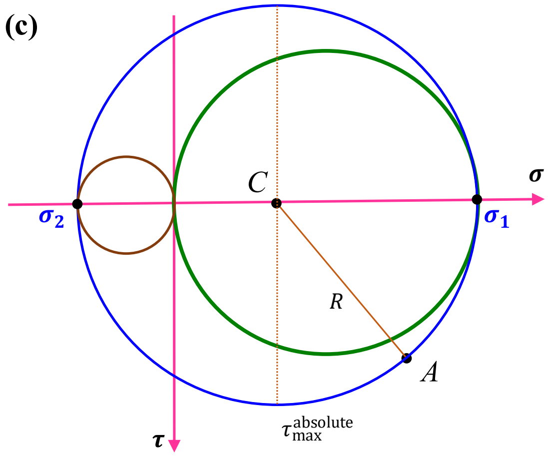
There are three possibilities to plot the absolute maximum shear stress:



For case (a): $\boxed{\sigma_1 = 2\tau_{\max}^{\text{abs}} = \mathbf{260\ MPa},\ \sigma_2 = \mathbf{0}}$

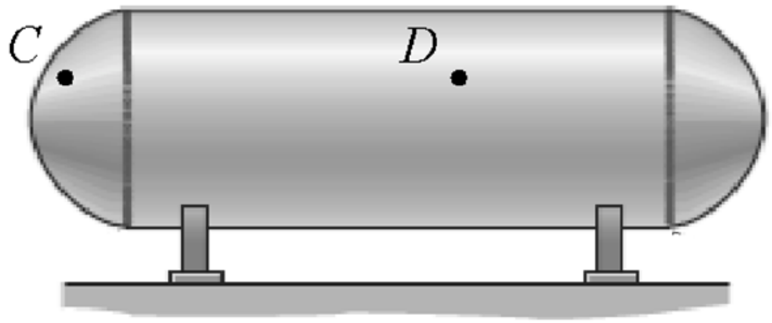
For case (b): $\boxed{\sigma_1 = 2R_C = \mathbf{190\ MPa},\ \sigma_2 = \sigma_1 - 2\tau_{\max}^{\text{abs}} = \mathbf{-70\ MPa}}$

(c)



For case (c): $R = \tau_{max}^{abs}$, $\boxed{\sigma_1 = \sigma_C + R = \mathbf{225\ MPa},\ \ \sigma_2 = \sigma_C - R = \mathbf{-35\ MPa}}$

***[Bonus Problem 2]** A pressurized steel tank has inner radius $r = 0.5$ m and wall thickness $t = 20$ mm. The steel has modulus of elasticity $E = 260$ GPa and Poisson's ratio $\nu = 0.3$. For a point D on the surface of the pressurized tank, if the absolute maximum shear strain is 5×10^{-4}, determine the internal pressure of the tank. (**10'** in total)



Solutions:

This is a plane stress problem. For a surface point, the normal stress in the hoop direction is twice as the normal stress in the longitudinal direction and the Mohr's circle can be plotted as shown. See page #5 in PPT Ch.9.4 for details.

The two principal stresses are:

$$\sigma_1 = \frac{pr}{t} \qquad \textbf{(1)}$$

$$\sigma_2 = \frac{pr}{2t} = \sigma_1/2 \qquad \textbf{(2)}$$

The radius of the Mohr's circle is $R = \frac{\sigma_1 - \sigma_2}{2} = \sigma_2/2$

The absolute maximum shear stress is:

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = \sigma_2 \qquad \textbf{(3)}$$

The shear modulus can be obtained by the material property relationship:

$$G = \frac{E}{2(1+\nu)} = \frac{260 \times 10^9}{2 \times (1+0.3)} = 100 \text{ GPa}$$

The absolute maximum shear stress can be obtained by using the absolute maximum shear strain and the generalized Hooke's law:

$$\tau_{\max}^{\text{abs}} = \gamma_{\max}^{\text{abs}} \cdot G$$

$$= 100 \times 10^9 \times 5 \times 10^{-4} = 5 \times 10^7 \text{ Pa}$$

Substituting into Eqs. (2) and (3) we have:

$$\sigma_2 = \tau_{\max}^{\text{abs}} = 50 \text{ MPa}$$

$$p = \frac{2\sigma_2 t}{r} = \frac{2 \times 50 \times 0.02}{0.5} = \boxed{\textbf{4 MPa}}$$